

numpy

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data visulization is visualize the data using data set like graph,bar,box etc... this kind of data had been represting the data visulization

```
pip install numpy
```

Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages (2.0.2)

```
import numpy as np
```

```
print(np.__version__)
```

2.0.2

```
a=np.array([1,2,3,4,5])
print(a)
```

[1 2 3 4 5]

```
b=np.array([[1,2,3],[4,5,6]])
print(b)
```

[[1 2 3]  
[4 5 6]]

```
c=np.array([[[1,2,3],[4,5,6]],[[7,8,9],[10,11,12]]])
print(c)
```

[[[ 1 2 3]  
[ 4 5 6]]  
  
[[ 7 8 9]  
[10 11 12]]]

```
d=np.full([3,3],6)
print(d)
```

[[6 6 6]  
[6 6 6]  
[6 6 6]]

```
e=np.zeros((3,3))
print(e)
```

[[0. 0. 0.]  
[0. 0. 0.]  
[0. 0. 0.]]

```
n=np.ones((3,3))
print(n)
```

[[1. 1. 1.]  
[1. 1. 1.]  
[1. 1. 1.]]

```
f=np.eye(3)
print(f)
```

[[1. 0. 0.]  
[0. 1. 0.]  
[0. 0. 1.]]

```
print(e,f,n,d)
```

[[0. 0. 0.]  
[0. 0. 0.]  
[0. 0. 0.]] [[1. 0. 0.]  
[0. 1. 0.]

```
[0. 0. 1.] [[1. 1. 1.]
[1. 1. 1.]
[1. 1. 1.]] [[6 6 6]
[6 6 6]
[6 6 6]]
```

```
f=np.arange(2,10,3)
print(f)
```

```
[2 5 8]
```

```
for i in range(2,10,3):
    print(i)
```

```
2
5
8
```

```
np.linspace(2,10,4)
```

```
array([ 2.          ,  4.66666667,  7.33333333, 10.          ])
```

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```
np.random.rand(3,3)
```

```
array([[0.86709689, 0.60893992, 0.65318284],
       [0.93411659, 0.32292964, 0.3686679 ],
       [0.83054314, 0.24202084, 0.62665682]])
```

```
np.random.randint(2,3)
```

```
2
```

```
np.random.randn(2,3)
```

```
array([[ -1.39099901,  1.90803335,  0.88354353],
       [-1.5270989 ,  1.30511557,  0.15063077]])
```

```
np.empty([2,3])
```

```
array([[1.19128795, 1.81105857, 0.063278 ],
       [0.70319233, 1.22553282, 0.3507649 ]])
```

```
arr=np.array(["a","b","c","d","e","f"])
print(arr.shape)
print(arr.ndim)
print(arr.size)
print(arr.dtype)
print(arr.itemsize)
print(arr.nbytes)
```

```
(1, 6)
2
6
<U1
4
24
```

```
a=np.array([1,2,3,4,5,6,7])
print(a[0])
print(a[2])
print(a[-1])
```

```
1
3
7
```

```
print(a[1:4])
print(a[:4])
print(a[1:6:2])
print(a[::-1])
```

```
[2 3 4]
[1 5]
[2 4 6]
[7 6 5 4 3 2 1]
```

```
m=np.array([[1,2,3,4],[5,6,34,2],])
print(m)
```

```
[[ 1  2  3  4]
 [ 5  6 34  2]]
```

```
print(m[0,1])
print(m[1,3])
print(m[0,2])
print(m[0,0])
```

```
2
2
3
1
```

```
print(m[0:])
print(m[1,:])
```

```
[[ 1  2  3  4]
 [ 5  6 34  2]]
[ 5  6 34  2]
```

```
print(m[:, 2])
```

```
[[1 2 3 4]]
```

```
print(m[1:2,1:2])
```

```
[[6]]
```

```
print(m)
#mask
l=m>10
print(l)
```

```
[[ 1  2  3  4]
 [ 5  6 34  2]]
[[False False False False]
 [False False  True False]]
```

```
#shothand
print(m[m>10])
print(m[1])
```

```
[34]
[34]
```

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